

$$\Rightarrow (q_0, aaaa bbbb, z_0) \xrightarrow[M]{*} (q_2, z_0)$$

Since the final state q_2 is reached stack is empty and the input symbol is consumed (input is ϵ). Thus the language is accepted by PDA.

Design of PDAs :

One basic design strategy for PDA is as follows

1. Understand the language properties for which the PDA has to be designed
2. Determine the state and alphabet set required
3. Identify the initial, accepting and dead states of PDA
4. Decide on the stack symbols required
5. Determine the initial stack symbol from the stack symbol set
6. For each state decide on the transition to be made for each character of the input string.
7. For each state transition, decide on the stack operations to be performed
8. Obtain the transition diagram and table
9. Test the PDA obtained on short strings

Example

Design a PDA to accept the language

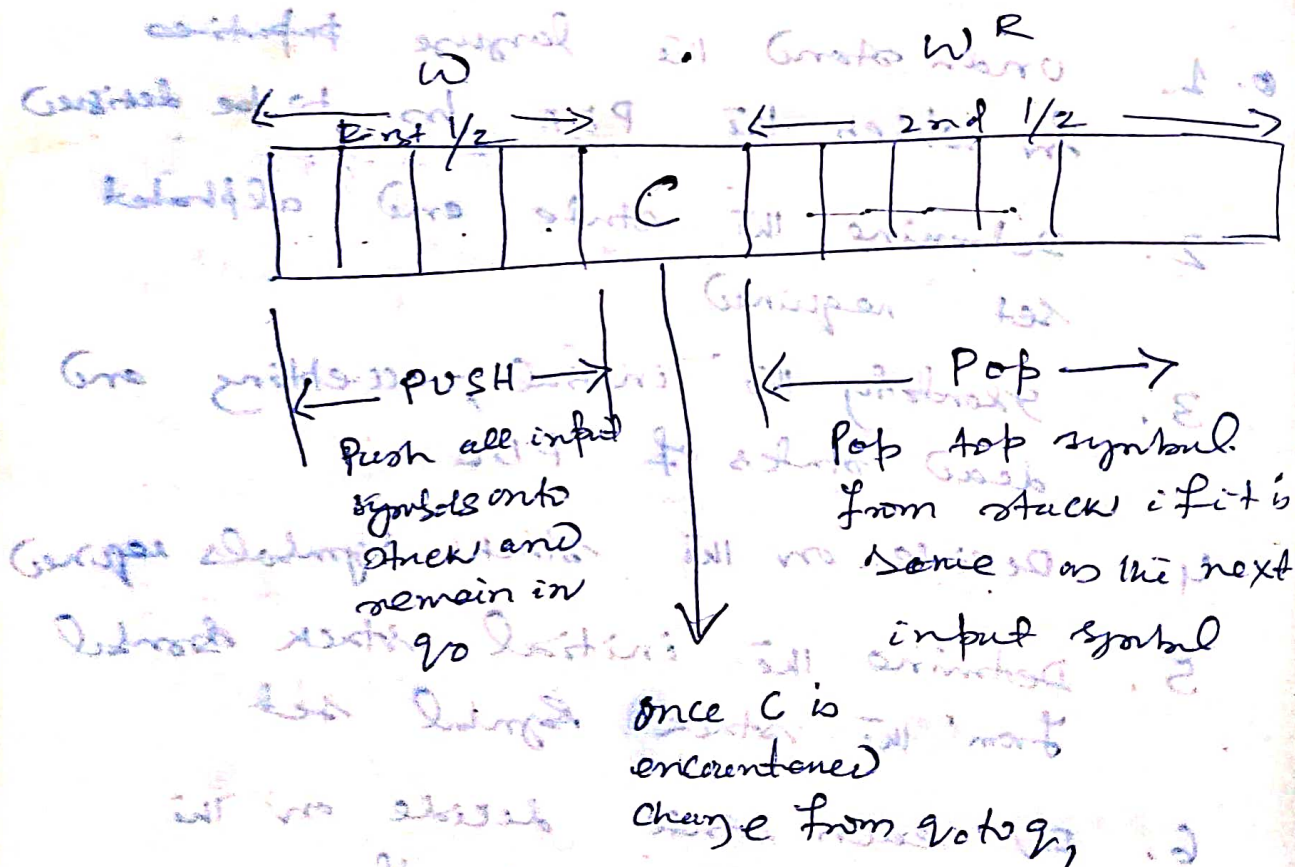
$$L = \{ w C w^R \mid w \in (0+1)^* \}$$

by an empty stack

or

Design a PDA that accepts the language of odd palindrome.

Design strategy.



Design view of PDA

Following are the transitions

i. Read each symbol of the input string and push onto stack before c is encountered.

a. $(q_0, \epsilon, Z_0, \text{nop}, q_0)$

b. $(q_0, 0, Z_0, \text{push}(0), q_0)$

c. $(q_0, 1, Z_0, \text{push}(1), q_0)$

d. $(q_0, 0, 0, \text{push}(0), q_0)$

e. $(q_0, 1, 1, \text{push}(1), q_0)$

f. $(q_0, 1, 0, \text{push}(1), q_0)$

g. $(q_0, 0, 1, \text{push}(0), q_0)$

ii) once c is encountered change q_1 without performing any operation.

h. $(q_0, c, 0, \text{nop}, q_1)$

i. $(q_0, c, 1, \text{nop}, q_1)$

j. $(q_0, c, Z_0, \text{nop}, q_1)$

iii) Read the next input symbol in the second half and pop from the stack if there is a match

k. $(q_1, 0, 0, \text{pop}, q_1)$

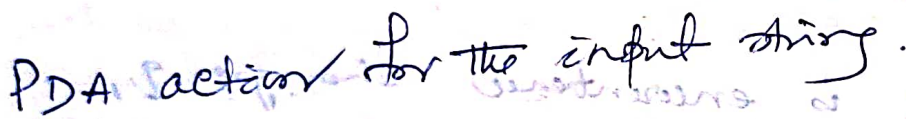
l. $(q_1, 1, 1, \text{pop}, q_1)$

iv. Once the stack is empty in the input

is ϵ , change to q_2

m. $(q_1, \epsilon, Z_0, \text{nop}, q_2)$

Transition diagrams


$$W = 001C \underline{100}$$

new input	Transition	Stack	New Stack
001C100	$(q_0, 0, 20, push(0), q_0)$	0 20	q_0
01C100	$(q_0, 0, 0, pop, q_0)$	00 20	q_0
1C100	$(q_0, 1, 0, push(1), q_0)$	100 20	q_0
C100	$(q_0, C, 1, not, q_1)$	100 20	q_1
100	$(q_1, 1, 1, pop, q_1)$	00 20	q_1
00	$(q_1, 0, 0, pop, q_1)$	0 20	q_1
0	$(q_1, 0, 0, pop, q_1)$	0 20	q_1
ε	$(q_1, ε, 20, not, q_2)$	20	q_2

a. To show that $w = 001C100$ is accepted by PDA is

ED

$$(q_0, 001C100, z_0) \vdash (q_0, 01C100, 0z_0)$$

$$\vdash (q_0, 1C100, 00z_0)$$

$$\vdash (q_0, C100, 100z_0)$$

$$\vdash (q_1, 100, 100z_0)$$

$$\vdash (q_1, 00, 00z_0)$$

$$\vdash (q_1, 0, 00z_0)$$

$$\vdash (q_1, \epsilon, z_0)$$

$$\vdash (q_2, z_0)$$

Since the final state is q_2 and the stack empty z_0 is reached it follows that w is accepted by PDA

b. To show that $01C00$

$$(q_0, 01C00, z_0) \vdash$$

$$\vdash (q_0, 1C00, 0z_0)$$

$$\vdash (q_0, C00, 10z_0)$$

$$\vdash (q_1, 00, 10z_0) \quad \times$$

$$\vdash$$

Δ_3 (There is no operation available for $(q_1, 0, 1)$ so it is rejected).